The Twin Identity, structurally: a multiset-tensor-product proof in the canonical σ_1-G1 basis

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Abstract

On April 29 the polynomial identity $A_{(3,2,1)}(q)+A_{(5,1)}(q) = A_{(3,1,1)}(q)\big(A_{(3,2)}(q)+A_{(4,1)}(q)\big)$ was verified by direct expansion. Here we lift it one structural rung. Working in the canonical σ_1-G1 basis $B_{2n} = \{q^c(1+q)^{2n-2c}\}_{c=0}^{n}$, each atom $A_\lambda(q)$ becomes a multiset M_λ on $\{0,1,\ldots,n\}$. Multiplication of σ_1-G1 polynomials acts on these multisets by convolution. The Twin Identity then upgrades to a clean multiset-level equality $M_{(3,2,1)} \uplus M_{(5,1)} = M_{(3,1,1)} * \big(M_{(3,2)} \uplus M_{(4,1)}\big)$, verified by tabulation. We are honest about what this is not yet: we have not exhibited an explicit bijection at the W-graph closed-path level. That remains open.

1 Setup

We work inside the σ_1-G1 framework developed in `2026-04-24-sigma1-G1.tex` (same directory). For each partition $\lambda \vdash n$, the σ_1-G1 atom $A_\lambda(q) \in \mathbb{Z}[q]$ is the closed-path generating polynomial of the W-graph of the irreducible S_n-module V_λ, with statistic c counting (recall the Apr 24 construction) the descent–ascent pairings along each closed path. We will not redevelop the construction here; the reader is referred to that note.

Definition 1 (Canonical σ_1-G1 basis)**.** For $n \geq 1$, let

$$B_{2n} \;=\; \left\{\, q^c(1+q)^{2n-2c} \,:\, c = 0,1,\ldots,n \,\right\} \;\subset\; \mathbb{Z}[q].$$

Each element of B_{2n} is a clean palindromic monomial in the two atoms q and $(1+q)^2$, of total q-degree $2n$ when read with the convention $\deg q = 1$, $\deg(1+q)^2 = 2$.

Lemma 2 (Uniqueness)**.** *B_{2n} is linearly independent over $\mathbb{Q}[q]$. Hence every polynomial in $\mathbb{Q}[q]$ supported in degrees $\leq 2n$ that lies in the $\mathbb{Q}$-span of B_{2n} admits a* unique *expansion $f(q) = \sum_{c=0}^{n} m_c \cdot q^c(1+q)^{2n-2c}$ with $m_c \in \mathbb{Q}$.*

Proof. Specialise $q = 0$ to recover m_0; subtract; divide by q; repeat. The recursion terminates and is well-defined. (Equivalently: the change-of-basis matrix from B_{2n} to the monomial basis $\{1, q, q^2, \ldots, q^{2n}\}$ is upper-triangular with nonzero diagonal.) $\square$

Definition 3 (Multiset of an atom)**.** Given an atom $A_\lambda(q)$ with $\lambda \vdash n$ admitting an expansion $A_\lambda(q) = \sum_{c=0}^{n} m_c^{(\lambda)} \cdot q^c(1+q)^{2n-2c}$, define

$$M_\lambda \;=\; \{\!\{\, (c, m_c^{(\lambda)}) \,:\, c = 0,\ldots,n,\; m_c^{(\lambda)} \neq 0 \,\}\!\}$$

as the multiset of (statistic, multiplicity) pairs. Equivalently M_λ is the function $c \mapsto m_c^{(\lambda)}$ on $\{0,\ldots,n\}$, regarded as a finitely supported $\mathbb{Z}_{\geq 0}$-valued measure.

By the Apr 24 framework, M_λ is precisely the multiset of closed-path c-statistics in the W-graph of V_λ.

2 The five shapes

The data we need are the canonical-basis decompositions of five specific atoms, two at S_6 and three at S_5:

$$A_{(3,2,1)}(q) = 1 \cdot (1+q)^6 + 9 \cdot q(1+q)^4 + 15 \cdot q^2(1+q)^2 + 2 \cdot q^3, \tag{1}$$

$$A_{(5,1)}(q) = 1 \cdot (1+q)^6 + 3 \cdot q(1+q)^4 + 9 \cdot q^2(1+q)^2 + 14 \cdot q^3, \tag{2}$$

$$A_{(3,1,1)}(q) = 1 \cdot (1+q)^2 + 2 \cdot q, \tag{3}$$

$$A_{(3,2)}(q) = 1 \cdot (1+q)^4 + 5 \cdot q(1+q)^2 + 3 \cdot q^2, \tag{4}$$

$$A_{(4,1)}(q) = 1 \cdot (1+q)^4 + 3 \cdot q(1+q)^2 + 5 \cdot q^2. \tag{5}$$

The decompositions (1)–(5) are unique by Lemma 2; the multiplicity vectors are

$$M_{(3,2,1)} = (1, 9, 15, 2), \quad M_{(5,1)} = (1, 3, 9, 14), \quad M_{(3,1,1)} = (1, 2),$$

$$M_{(3,2)} = (1, 5, 3), \quad M_{(4,1)} = (1, 3, 5).$$

Already a surprise jumps out:

Remark 4 (The reflected $n = 5$ pair)**.** The vectors $M_{(3,2)} = (1, 5, 3)$ and $M_{(4,1)} = (1, 3, 5)$ are reflections of each other under $c \mapsto 2 - c$ on B_4. So $A_{(3,2)}$ and $A_{(4,1)}$ are related by an involution on σ_1-G1 multisets. Note that this is *not* plain conjugation of partitions: $(3, 2)' = (2, 2, 1) \neq (4, 1)$. Whatever this involution is at the W-graph level, it is finer than $\lambda \leftrightarrow \lambda'$.

3 The convolution lemma

Lemma 5 (B_{2n} is preserved under multiplication)**.** *For all $a, b \geq 0$ and all $0 \leq c_a \leq a$, $0 \leq c_b \leq b$,*

$$\left[q^{c_a}(1+q)^{2a-2c_a}\right] \cdot \left[q^{c_b}(1+q)^{2b-2c_b}\right] \;=\; q^{c_a+c_b}(1+q)^{2(a+b)-2(c_a+c_b)} \;\in\; B_{2(a+b)}.$$

Hence multiplication of σ_1-G1 polynomials acts on multisets by convolution on the c-coordinate: if $f = \sum_a m_a \cdot q^a(1+q)^{2A-2a}$ and $g = \sum_b m'_b \cdot q^b(1+q)^{2B-2b}$, then

$$fg \;=\; \sum_{c=0}^{A+B} \Big(\sum_{a+b=c} m_a\, m'_b \Big) \cdot q^c(1+q)^{2(A+B)-2c}.$$

Proof. The first claim is the calculation

$$q^{c_a}(1+q)^{2a-2c_a} \cdot q^{c_b}(1+q)^{2b-2c_b} = q^{c_a+c_b}(1+q)^{(2a-2c_a)+(2b-2c_b)} = q^{c_a+c_b}(1+q)^{2(a+b)-2(c_a+c_b)}.$$

Since $0 \leq c_a + c_b \leq a + b$, the result lies in $B_{2(a+b)}$. The second claim is bilinearity. □

This is the structural content. The basis B_{2n} is closed under products, and the product law on the basis is *additive in c*. So in the multiset language, polynomial multiplication **is** discrete convolution.

4 The structural twin identity

Write $\uplus$ for multiset union (pointwise sum of multiplicity functions) and $*$ for convolution.

Theorem 6 (Structural twin identity)**.**

$$M_{(3,2,1)} \uplus M_{(5,1)} = M_{(3,1,1)} * \big(M_{(3,2)} \uplus M_{(4,1)}\big).$$

Proof. Direct computation. The left-hand multiset, on $\{0,1,2,3\}$, is

$$M_{(3,2,1)} \uplus M_{(5,1)} = (1,9,15,2) + (1,3,9,14) = (2,\ 12,\ 24,\ 16).$$

For the right-hand side, first take the inner union on $\{0,1,2\}$:

$$M_{(3,2)} \uplus M_{(4,1)} = (1,5,3) + (1,3,5) = (2,\ 8,\ 8).$$

Now convolve with $M_{(3,1,1)} = (1,2)$:

$$\begin{aligned}(1,2)*(2,8,8) &= \big(1\cdot 2,\ \ 1\cdot 8 + 2\cdot 2,\ \ 1\cdot 8 + 2\cdot 8,\ \ 2\cdot 8\big)\\ &= (2,\ 12,\ 24,\ 16).\end{aligned}$$

The two multisets agree. $\square$

It is illuminating to verify the convolution at the polynomial level, since this is what Lemma 5 tells us we are computing:

$$\begin{aligned}A_{(3,1,1)}\big(A_{(3,2)} + A_{(4,1)}\big) &= \big((1+q)^2 + 2q\big)\cdot\big(2(1+q)^4 + 8q(1+q)^2 + 8q^2\big)\\ &= 2(1+q)^6 + 8q(1+q)^4 + 8q^2(1+q)^2\\ &\quad + 4q(1+q)^4 + 16q^2(1+q)^2 + 16q^3\\ &= 2(1+q)^6 + 12\,q(1+q)^4 + 24\,q^2(1+q)^2 + 16\,q^3.\end{aligned}$$

Each line stays inside the canonical basis: $(1+q)^2\cdot(1+q)^{2k} = (1+q)^{2k+2}$ slides c up by 0, while $q\cdot(1+q)^{2k}$ slides c up by 1 (and $q\cdot q\cdot(1+q)^{2k} = q^2(1+q)^{2k}$ slides by 2). No basis element ever splits into a sum of two; the multiplication preserves B_{2n+2}.

Corollary 7 (Theorem A, the polynomial twin identity)**.**

$$A_{(3,2,1)}(q) + A_{(5,1)}(q) = A_{(3,1,1)}(q)\cdot\big(A_{(3,2)}(q) + A_{(4,1)}(q)\big).$$

Proof. Apply Lemma 5 to translate the multiset identity of Theorem 6 into polynomials. (The polynomial identity was already verified directly on April 29; Theorem 6 is the structural lift.) $\square$

5 Remarks and open questions

Remark 8 (The $(3,2)\leftrightarrow(4,1)$ involution)**.** $M_{(3,2)} = (1,5,3)$ and $M_{(4,1)} = (1,3,5)$ swap under the reflection $c \mapsto 2-c$ on B_4. As noted, this is *not* the conjugation involution on partitions. It is conceivably the conjugation involution composed with the sign character, or a Mullineux-like map, or something genuinely new at the level of σ_1-G1 paths. Pinning it down is one of the cleanest n=5 questions.

Remark 9 (What is missing: a path-level bijection)**.** The structural identity says: the multiset of closed-path c-statistics on the LHS equals that on the RHS. It does *not* construct a bijection

$$\mathrm{paths}(V_{(3,2,1)}) \ \sqcup\ \mathrm{paths}(V_{(5,1)}) \ \xrightarrow{\sim}\ \mathrm{paths}(V_{(3,1,1)}) \ \times\ \big(\mathrm{paths}(V_{(3,2)}) \sqcup \mathrm{paths}(V_{(4,1)})\big)$$

preserving the c-statistic. Such a bijection — presumably built from $S_6 \downarrow S_5 \times S_1$ branching, or perhaps from the LLT framework — would upgrade Theorem 6 from a multiset coincidence to a genuine structural-tensor-product theorem. This is the next rung. Open.

Remark 10 (Relation to the 2q-perturbation rule (Theorem C))**.** The 2q-perturbation rule, conjecturally controlling the orphan paths, refines the multiset identity by tracking a distinguished sub-multiset on each side. We expect Theorem 6 to factor through Theorem C: the 2q-perturbation isolates the orphan contribution, and the remainder convolves cleanly. Future work.

Computational verification

All five expansions (1)–(5), the union $(1, 9, 15, 2) + (1, 3, 9, 14) = (2, 12, 24, 16)$, the union $(1, 5, 3) + (1, 3, 5) = (2, 8, 8)$, and the convolution $(1, 2) * (2, 8, 8) = (2, 12, 24, 16)$ were verified May 6, 2026 in `~/projects/scratch/2026-05-06-twin-wgraph/`.